

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/Marks
2025 – Present	M.Tech, Computer Science and Engineering	Indian Institute of Technology, Kanpur	8.95/10
2020 – 2024	B.Tech, Computer Science and Engineering	Vardhaman College of Engineering, Hyderabad	8.8/10
2018 – 2020	Intermediate (Class XII)	Sri Chaitanya Junior College, Hyderabad	948/1000
2018	Class X (SSC)	SRK High School, Siddipet	9.0/10

M Tech Thesis

- **Ambiguity in Coreference Resolution for Large Language Models** (Ongoing) (May'26- Present)
Mentor: **Prof. Arnab Bhattacharya**, Department of Computer Science and Engineering, IIT Kanpur.
 - Developing a **Hindi-language benchmark** containing ambiguous, non-ambiguous, and cue-dependent coreference instances to evaluate LLM reasoning over discourse context.
 - Designed a **multiple-choice evaluation framework** to measure LLM ability to resolve ambiguous references using contextual information, coreference chains, and discourse cues.
 - Evaluating how **explicit coreference and discourse information** affects LLM accuracy in resolving ambiguous references across varying contextual conditions.

Projects

- **CodeGrep: Code-Aware RAG** | Self Project 🐙 | tree-sitter, sqlite-vec, FTS5, sentence-transformers (Aug'26)
 - Built **AST-based chunking** with tree-sitter that splits Python at function and class boundaries and prepends file-path, scope, and import context to every chunk; a three-tier fallback (AST → statement → line) held **11,916 chunks** from numpy inside budget with zero overflow in **1.4 s**.
 - Implemented **hybrid retrieval** fusing dense vectors (sqlite-vec) with BM25 (SQLite FTS5) via **Reciprocal Rank Fusion**, fusing on ranks rather than scores since cosine and BM25 occupy incomparable scales; hybrid reached **1.000 MRR** on exact-identifier and documentation queries versus 0.900 and 0.778 for dense alone.
 - Built a 45-question evaluation harness (recall@k, MRR, nDCG) and **reported a negative result**: cross-encoder reranking cost **40× latency** (611 ms vs 16 ms) for lower MRR, because dense recall@3 was already **0.97**, leaving no ranking headroom to recover.
 - Calibrated **abstention** against 15 deliberately unanswerable questions including domain-plausible hard negatives, finding a **3.2-point score gap** that separates in-scope from out-of-scope queries: **30/30** answerable answered, **15/15** unanswerable declined.
- **CodeSense AI: Local-First Code Review Assistant** | Self Project 🐙 | FastAPI, Qwen2.5-Coder, Ollama (July'26)
 - Built a local-first code review tool using **Qwen2.5-Coder:7B** via Ollama, exposing style review, complexity analysis, and edge-case generation through FastAPI with a React Monaco code-editor frontend, SQLite persistence, and Docker Compose.
 - Designed a **swappable provider abstraction** so adding a cloud backend (Groq) required one new file and one factory entry, without modifying routes, services, schemas, or frontend.
 - Hardened structured LLM output using mandatory-evidence response schemas, a validator that drops self-refuting claims, JSON parse-and-retry, and temperature fixed at 0.1 after default sampling produced scores of 89/70/30 on identical input.
 - Benchmarked latency across models and deployments: **2.2–6.3 s** locally on RTX 5060 versus **0.7–1.9 s** server-side on Groq, separating network from inference latency.
- **SmartSMS: On-Device SMS Manager (Android)** | Self Project 🐙 | ReactNative(CLI), TFLite, Node.js (May'26)
 - Fine-tuned DistilBERT for 6-class SMS classification, deployed on-device via TFLite with a custom Kotlin WordPiece tokenizer, achieving **100% OTP recall** while ensuring no user data leaves the device.
 - Benchmarked DistilBERT against MobileBERT; ablation showed removing hard negatives dropped adversarial accuracy from **100% to ~47%**.
 - Diagnosed INT8 quantization accuracy loss and shipped FP32 after evaluation; built an active-learning pipeline for continuous retraining.

- Built the Node.js/Express backend with JWT refresh-token rotation, Google OAuth, multi-device synchronization, Dockerized deployment, and rate-limiting.
 - Developed the React Native Android app with custom Kotlin native modules, MMKV storage, WorkManager background processing, and default-SMS-role support.
- **RescueSeg: Hybrid-Attention Transformer for Segmentation** | Course Project(CS776) 🌀 | PyTorch (Apr'26)
 - Co-developed a hybrid encoder combining Overlapping Cross Attention and Global Sparse Attention through a learnable fusion gate, with FPN, Attention U-Net decoder, and boundary refinement.
 - Achieved **77.35% mIoU** at **44 ms** inference on RescueNet with 11 classes, outperforming a Segmenter baseline under identical settings.
 - Ran a 3-variant ablation showing boundary refinement and auxiliary supervision prevent NaN divergence during epochs 14–20, improving optimization stability.
 - **Distributed 3D Stencil Solver with MPI Halo Exchange** | Course Assignment(CS633) | C, MPI (Apr'26)
 - Built a distributed d -point stencil solver over a $p_x \times p_y \times p_z$ Cartesian process grid with six-neighbour ghost-cell halo exchange.
 - Eliminated pack/unpack overhead by communicating non-contiguous halo faces through MPI derived datatypes.
 - Overlapped communication with computation via non-blocking sends/receives, computing interior points during message transit; replaced array copies with pointer swapping.
 - Benchmarked on **PARAM Rudra** across **32–96 processes**, measuring a consistent **1.17–1.20 $\times$ speedup** at 240^3 with communication overhead limiting gains at smaller sizes.
 - **Music Genre Classification from Audio Spectrograms** | Course Assignment(CS776) | Python, PyTorch (Feb'26)
 - Implemented CNN primitives from scratch instead of built-in layers: 2D convolution via unfold sliding windows and einsum contraction, plus max pooling, channel normalization, and the classifier head.
 - Built a ResNet-v2 pre-activation bottleneck network with **380,920 parameters** over 3-channel inputs stacking log-mel spectrogram, MFCC, and chroma features per clip.
 - Scored **0.92993 macro F1** on the competition leaderboard across **16 classes** with 14,550 held-out test samples; validation macro F1 differed by only 0.0002.
 - Owned training methodology including optimizer comparison (SGD / AdamW / Adam), batch-size generalization analysis, **OneCycleLR** scheduling, label smoothing, and early stopping.
 - Ran ablations on input representation, resolution ($124 \times 92 \rightarrow 248 \times 184$), and per-feature normalization.
 - **Text-to-SQL via Dynamic Few-Shot Prompting** | Course Project(CS787) 🌀 | LLMs, SentenceTransformers (Nov'25)
 - Built a **retrieval-augmented** pipeline that embeds each question using SentenceTransformer (all-MiniLM-L6-v2) and retrieves the top-K most cosine-similar solved examples from a held-out knowledge base to construct the prompt.
 - Ran a controlled comparison of four prompting strategies across two models (Gemini 2.5 Flash, Llama 3.1 8B) and two benchmarks (Spider, BIRD), evaluated by execution accuracy on a 200-sample subset per dataset.
 - Raised Gemini execution accuracy on **BIRD from 33.0% to 52.5%** and on **Spider from 61.0% to 74.5%** over the zero-shot baseline; Llama improved from 8.0% to 32.5% on BIRD.
 - Swept retrieval depth $K = 1-5$, finding gains saturate near $K = 3-4$ and attributing the ceiling to context saturation and BIRD's implicit foreign keys and non-standard column naming.
 - **Kartly: E-Commerce Web Application** | Self Project 🌀 | React, Redux Toolkit, Firebase (Oct'23)
 - Built a single-page storefront with Redux Toolkit as the centralized state manager, isolating all side effects in createAsyncThunk layers so components stayed presentational and never called Firebase or the product API directly.
 - Implemented per-user cart and order persistence on Cloud Firestore with realtime onSnapshot listeners, pushing changes into Redux so state stays in sync across tabs and devices without polling.
 - Designed a chained filter pipeline (category $\rightarrow$ price range $\rightarrow$ search) over the catalogue, alongside Firebase Auth email/password sign-in, protected routes, and a router-level error boundary via createBrowserRouter.

Technical Skills

- **Languages:** Python, C, C++, SQL, JavaScript
- **Machine Learning:** PyTorch, Hugging Face Transformers, TensorFlow Lite, NumPy, Pandas, Quantization
- **LLM Systems:** Retrieval-Augmented Generation (RAG), Prompt Engineering
- **Parallel & HPC:** MPI (non-blocking communication, derived datatypes), SLURM, Scaling Analysis, Performance Benchmarking
- **Backend:** FastAPI, Node.js, Express, REST APIs, JWT, OAuth
- **Frontend / Mobile:** React, React Native (CLI), Redux Toolkit, HTML, CSS3
- **Databases / Storage:** MongoDB (Mongoose), MySQL
- **Others:** Docker, Git, GitHub, Postman, Hugging Face Hub

Relevant Courses

Design & Analysis of Algorithms Deep Learning for Comp Vision Parallel Computing	Intro to Machine Learning Generative Artificial Intelligence Algorithmic Information Theory
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